# SQL query-based questions focusing on date functions for SQL Server:

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-- Create the Orders Table

	CREATE TABLE Orders (
		OrderID INT PRIMARY KEY,
		OrderDate DATE,
		CustomerID INT,
		Amount DECIMAL(10, 2));

-- 2. Insert Sample Data

	INSERT INTO Orders (OrderID, OrderDate, CustomerID, Amount) VALUES
	(1, '2024-08-15', 101, 250.00),
	(2, '2024-08-20', 102, 150.00),
	(3, '2024-07-25', 103, 200.00),
	(4, '2024-06-10', 104, 300.00),
	(5, '2024-06-15', 105, 450.00),
	(6, '2024-09-05', 106, 120.00),
	(7, '2024-05-20', 107, 180.00),
	(8, '2024-04-10', 108, 210.00),
	(9, '2024-03-15', 109, 320.00),
	(10, '2024-02-10', 110, 270.00);

select * from orders;

OUTPUT:
OrderID     OrderDate        CustomerID  Amount
----------- ---------------- ----------- ------------
          1       2024-08-15         101       250.00
          2       2024-08-20         102       150.00
          3       2024-07-25         103       200.00
          4       2024-06-10         104       300.00
          5       2024-06-15         105       450.00
          6       2024-09-05         106       120.00
          7       2024-05-20         107       180.00
          8       2024-04-10         108       210.00
          9       2024-03-15         109       320.00
         10       2024-02-10         110       270.00

#############################################################################################################
1. Question: Retrieve records from the Orders table where the order date is within the last 30 days.
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Sample Table:
Orders (OrderID, OrderDate, CustomerID, Amount)
Query:


	SELECT OrderID, OrderDate, CustomerID, Amount
	FROM Orders
	WHERE OrderDate >= DATEADD(DAY, -30, GETDATE());

OUTPUT:
OrderID     OrderDate        CustomerID  Amount
----------- ---------------- ----------- ------------
          1       2024-08-15         101       250.00
          2       2024-08-20         102       150.00
          6       2024-09-05         106       120.00

OUTPUT:
Explanation:
GETDATE() returns the current date and time.
DATEADD(DAY, -30, GETDATE()) calculates the date 30 days before today.
The WHERE clause filters records where OrderDate is on or after this calculated date.
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#########################################################################################################
2. Question: Find the number of orders placed in each month of the current year.
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Sample Table:
Orders (OrderID, OrderDate, CustomerID, Amount)
Query:

	SELECT
		YEAR(OrderDate) AS Year,
		MONTH(OrderDate) AS Month,
		COUNT(OrderID) AS NumberOfOrders
	FROM Orders
	WHERE YEAR(OrderDate) = YEAR(GETDATE())
	GROUP BY YEAR(OrderDate), MONTH(OrderDate)
	ORDER BY Month;

OUTPUT:
Year        Month       NumberOfOrders
----------- ----------- --------------
	2024           2              1
	2024           3              1
	2024           4              1
	2024           5              1
	2024           6              2
	2024           7              1
	2024           8              2
	2024           9              1

Explanation:

YEAR(GETDATE()) gets the current year.
YEAR(OrderDate) and MONTH(OrderDate) extract the year and month from OrderDate.
COUNT(OrderID) counts the number of orders for each month.
GROUP BY groups the results by year and month.
ORDER BY Month sorts the results by month.


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3. Question: Calculate the average number of days between order dates for the Orders table.
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Sample Table:
Orders (OrderID, OrderDate, CustomerID, Amount)
Query:


	WITH DateDiffs AS (
		SELECT
			DATEDIFF(DAY, LAG(OrderDate) OVER (ORDER BY OrderDate), OrderDate) AS DaysBetween
		FROM Orders
	)
	SELECT AVG(DaysBetween) AS AverageDaysBetween
	FROM DateDiffs
	WHERE DaysBetween IS NOT NULL;

OUTPUT:
AverageDaysBetween
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                23

Explanation:

LAG(OrderDate) OVER (ORDER BY OrderDate) gets the previous order date in the sequence.
DATEDIFF(DAY, ..., OrderDate) calculates the difference in days between the current and previous order
date.
WITH DateDiffs AS (...) creates a common table expression (CTE) to calculate these differences.
AVG(DaysBetween) computes the average number of days between orders.
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#########################################################################################################
4. Question: Determine the first and last day of the current quarter.
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Sample Table:
Orders (OrderID, OrderDate, CustomerID, Amount)
Query:


        DECLARE @CurrentDate DATE = GETDATE();
        DECLARE @QuarterStart DATE;
        DECLARE @QuarterEnd DATE;

        -- Determine the start of the current quarter
        SET @QuarterStart = DATEADD(QUARTER, DATEDIFF(QUARTER, 0, @CurrentDate), 0);

        -- Determine the end of the current quarter
        SET @QuarterEnd = DATEADD(DAY, -1, DATEADD(QUARTER, 1, @QuarterStart));

        SELECT @QuarterStart AS QuarterStart, @QuarterEnd AS QuarterEnd;

OUTPUT:
QuarterStart     QuarterEnd
---------------- ----------------
      2024-07-01       2024-09-30

Explanation:

DATEADD(QUARTER, DATEDIFF(QUARTER, 0, @CurrentDate), 0) calculates the start date of the current quarter.
DATEADD(QUARTER, 1, @QuarterStart) calculates the start of the next quarter.
DATEADD(DAY, -1, ...) subtracts one day to get the last day of the current quarter.



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5. Question: Find the day of the week and week number for each order in the Orders table.
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Sample Table:
Orders (OrderID, OrderDate, CustomerID, Amount)
Query:

        SELECT
               OrderID,
               OrderDate,
               DATENAME(WEEKDAY, OrderDate) AS DayOfWeek,
               DATEPART(WEEK, OrderDate) AS WeekNumber
        FROM Orders
        ORDER BY OrderDate;

OUTPUT:
OrderID     OrderDate        DayOfWeek                      WeekNumber
----------- ---------------- ------------------------------ -----------
         10       2024-02-10 Saturday                                6
          9       2024-03-15 Friday                                 11
          8       2024-04-10 Wednesday                              15
          7       2024-05-20 Monday                                 21
          4       2024-06-10 Monday                                 24
          5       2024-06-15 Saturday                               24
          3       2024-07-25 Thursday                               30
          1       2024-08-15 Thursday                               33
          2       2024-08-20 Tuesday                                34
          6       2024-09-05 Thursday                               36

Explanation:

DATENAME(WEEKDAY, OrderDate) retrieves the name of the day of the week (e.g., 'Monday').
DATEPART(WEEK, OrderDate) gets the week number of the year for OrderDate.
ORDER BY OrderDate sorts the results by order date.
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